Sepehr Rezaee

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Education

Shahid Beheshti University, BS. in Computer Sciences 2021 – 2025

- **Interests:** Deep Learning, Computer Vision, Generative AI, and AI Safety

Allameh Tabatabaei (Advanced) High School, Math Diploma 2019 – 2021

- GPA: 3.87/4.0

Experience

Research Assistant, Robust and Interpretable Machine Learning Lab – Sharif University of Technology, Tehran 2024 – Present

- Authored and co-authored 3 papers submitted to NeurIPS 2024, focusing on enhancing model reliability and security in machine learning.
- Developed and implemented 3 robust machine learning pipelines, increasing model reliability by adversarial conditions.
- Collaborated with a multidisciplinary team of 10 members to integrate machine learning solutions into 3 real-world applications(Autonomous Driving, Face Detection, Diagnosing Disease), improving operational efficiency.
- Presented research findings at 2 international conferences, elevating the lab's visibility and fostering academic collaborations.

Research Assistant, Artificial Intelligence and Scientific Computing Lab – Shahid Beheshti University, Tehran 2023 – Present

- Co-authored 2 underereview & 1 published research papers, including:
  - *Physics-Informed Lane-Emden Solvers Using Lynx-Net: Implementing Radial Basis Functions in Kolmogorov Representation*
  - *Leveraging Physics-Informed Convolutional Neural Networks (PICNNs) to Solve Linear and Non-linear Fokker-Planck Equations (FPEs)*
  - *Comparison of Pre-training and Classification Models for Early Detection of Alzheimer's Disease Using Magnetic Resonance Imaging*
- Modeled disease progression using differential equations, enhancing the understanding of biological mechanisms.
- Employed Physics-Informed Neural Networks (PINNs), increasing model accuracy through the integration of physical laws.

AI Specializing In Applied LLMs, Propty 2024 – Present

- Led the development of an AI-powered chatbot and home finder system using open-source LLMs and Retrieval-Augmented Generation (RAG) to enhance real estate customer service.
- Engineered a scalable backend infrastructure with FastAPI, PostgreSQL, and MongoDB, ensuring high reliability and performance for real-time interactions.
- Integrated and fine-tuned LLMs (GPT-Neo, GPT-J, LLaMA) for domain-specific tasks, significantly improving the accuracy and contextual relevance of chatbot responses.
- Successfully deployed the solution with Docker and Kubernetes, achieving seamless scalability and handling a large volume of concurrent user requests.
- Implemented real-time monitoring and continuous improvement strategies using Prometheus, Grafana, and the ELK stack, ensuring system stability and performance.

Deep Learning and Neuroscience Intern Researcher, Institute for Research in Fundamental Sciences (IPM) – Tehran 2023 – 2024

- Conducted comprehensive M/EEG data analysis utilizing advanced deep learning techniques to decode neural signals.
- Developed and optimized neural network architectures for improved signal processing and feature extraction.
- Collaborated with neuroscientists to interpret data results and contribute to the understanding of brain functionalities.
- Assisted in the preparation of research manuscripts and presentations for academic dissemination.

Publications

Scanning Trojaned Models Using Out-of-Distribution Samples Accepted to NeurIPS 2024

Hossein Mirzaei, Ali Ansari*, Bahar Dibaei Nia*, Mojtaba Nafez†, Moein Madadi†, **Sepehr Rezaee**†, Zeinab Sadat Taghavi, Arad Maleki, Kian Shamsaie, Mahdi Hajialilue, Jafar Habibi, Mohammad Sabokrou, Mohammad Hossein Rohban

A Contrastive teacher-student framework for novelty detection under style shifts Submitted to ICLR 2025

Hossein Mirzaei, Mojtaba Nafez, Moein Madadi, Arad Maleki, Mahdi Hajialilue, Zeinab Sadat Taghavi, **Sepehr Rezaee**, Ali Ansari, Bahar Dibaei Nia, Kian Shamsaie, Mohammadreza Salehi, Jafar Habibi, Mackenzie W Mathis, Mahdieh Soleymani Baghshah, Mohammad Sabokrou, Mohammad Hossein Rohban

Comparison of pre-training and classification models for early detection of Alzheimer's disease using magnetic resonance imaging Accepted in ICCCCC 2023 — 2023

AH Karami, **S Rezaee**, E Mirzabeigi, K Parand

Hierarchical Clustering Algorithms,Chapter of Unsupervised Algorithms: Clustering (with Implementation) Aarvan Publications — 2022

Kourosh Parand, **Sepehr Rezaee**, et al.

Selected Projects

AI Model Security: Enhancing Robustness Against Backdoors and Trojans — 2024

- Developed methods to detect and mitigate backdoors in machine learning models, enhancing AI deployment security.
- Engineered algorithms using statistical analysis and pattern recognition, improving trojan detection rates.
- Contributed to NeurIPS 2024 publications, advancing the field of AI model security.
- **Tools Used:** Python, PyTorch, Scikit-learn, LaTeX

Developed an AI-based application for early detection of Alzheimer's disease. — 2023-2024

- Designed and implemented a customized multi-modal model integrating biomedical and MRI datasets.
- Enhanced diagnostic accuracy through advanced machine learning techniques with VLMs.
- **Tools Used:** PyTorch, Hugging Face, OpenCV

Physics-Informed Neural Networks for Disease Progression Modeling — 2023

- Created a Physics-Informed Neural Network integrating differential equations to accurately predict disease progression.
- Utilized clinical datasets and validated models with patient data, achieving higher accuracy than traditional methods.
- Published findings in peer-reviewed journals, contributing to AI-based healthcare innovations.
- **Tools Used:** PyTorch, NumPy, SciPy, Pandas

AI-Driven M/EEG Data Analysis for Neuroscience Research — 2022

- Applied deep learning techniques to decode M/EEG signals, uncovering neural mechanisms.
- Streamlined data workflows by automating preprocessing and artifact removal, enhancing analysis efficiency.
- Facilitated insights into brain connectivity, supporting high-impact neuroscience research publications.
- **Tools Used:** MNE-Python, PyTorch, NumPy, Pandas

Teaching Assistant

Advanced Programming Head Teaching Assistant, Shahid Beheshti University, Tehran — 2024 – Present

Data Mining and Analysis Head Teaching Assistant, Shahid Beheshti University, Tehran — 2023

Basic Programming Teaching Assistant, Shahid Beheshti University, Tehran — 2022

Assistant Teacher and Mentor — 2022 – 2023

- Applications of Data Science and Artificial Intelligence in the Petrochemical Industry, the Water Industry & the Electricity Industry

Selected Courses

Courses: Foundations of Data Science (A^+, 1st), Data Mining (A^+, 1st), Advanced Data Mining (A^+, 1st), Foundation of Numerical Analysis (A^+, 1st), Non-Linear Optimization (A^+, 1st), Partial Differential Equations (A^+, 1st), Electromagnetics (A^+, 1st), Neural Network (A^+, 3rd), Foundation of Logic and Set Theory (A^+, 3rd), Principles of Operating Systems (A^+, 2nd), Foundations of Machine Learning (A^+, 2nd), Elements of Probability (A, 4th), Data Structures & Algorithms (A, 5th)

Skills

Programming Languages: Python, C++, C, MATLAB, C# & Java

Python Frameworks & Libraries: PyTorch, TensorFlow, OpenCV, MNE-Python, NumPy, SciPy, Matplotlib, Scikit-Learn, NiPype, FastAPI, Django, Django REST Framework, Selenium

Other Tools and Technologies: JAX, PostgreSQL, NoSQL, MongoDB, Kotlin, , Git, Docker, Linux, Bootstrap

Interpersonal Skills: Problem Solving, Team Working

Languages: Fluent in Persian (speaking, reading, and writing), English (Professional working proficiency)

Reference Contacts

Prof. Kourosh Parand - k_parand@sbu.ac.ir
Prof. Mohammad Hossein Rohban - rohban@sharif.edu
Prof. Mohammad Sabokrou - mohammad.sabokrou@oist.jp